

Life cycle archive

Teacher guide and worked answers

Describe the life cycles of a mammal, an amphibian, an insect and a bird using named examples.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsg/inputs/GROW SOW 2026-27.xlsx | GROW Weekly - Spring | C37

Describe the life cycles of mammals, amphibians, insects and birds.

Directly develops this stated outcome with secondary-age tasks at an upper-KS2 conceptual level. Support and stretch are flexible; this exemplar is not a claim of qualification approval.

Prior learning

Recognise rabbits, frogs, butterflies and birds as animals.

Know that offspring grow and that living things reproduce.

Success evidence

Order stages for four named animal examples.

Use growth, development and reproduction to explain arrows.

Use evidence to correct an over-general claim.

Equipment

Shared evidence page, one route page and exit page; pencil; optional paper stage labels and reusable arrows.

Preparation

Print pages 1 and 5 plus one selected response page.

Read the named-species evidence before discussing whole animal groups. Prepare movable stage words if pupils benefit from sorting.

Practical care

Use paper evidence only. No animals, spawn, nests or eggs need to be collected or handled.

Keep reproduction discussion focused on the named wildlife examples; no personal or family disclosure is required.

Ready to teach alternative

All evidence and stage words are supplied. Pupils can point, sequence labels or explain aloud instead of drawing.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	The wildlife archive is mixed up
2	Retrieval	2-5	What changes as animals grow?
3	I do	5-13	I explain each frog arrow
4	I do	Within stage	I check for a pupal stage
5	I do	Within stage	I keep the animal name in the claim
6	We do	13-21	Repair three archive labels
7	We do	Within stage	Recover the missing stage
8	We do	Within stage	Challenge the word all
9	Hinge	21-24	What does the returning arrow mean?
10	You do	24-34	Build a reliable display
11	You do	Within stage	Check what your arrows say
12	Review	34-38	An audience checks the display
13	Review	Within stage	A smaller animal is not always an adult
14	Exit	38-40	Three final archive checks
15	Exit	Within stage	Choose a useful next step

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening The wildlife archive is mixed up

Introduce a paper museum challenge. Pupils may work alone or with a partner. The goal is describing life cycles, not recalling precise ages or drawing realistic animals.

2 Retrieval What changes as animals grow?

R1 young produced by parents. R2 no; it still develops and matures. R3 no. Accept everyday words, then supply offspring and mature.

3 I do I explain each frog arrow

Think aloud: I follow the same individual from egg through tadpole and froglet to adult. At reproduction the arrow now links parents to offspring. I do not say the adult changes back into an egg. The timing and shapes are simplified.

4 I do I check for a pupal stage

Shared I do time. Trace egg to larva to pupa to adult. Model the explanation: the caterpillar does not become a different species; its body develops. Use chrysalis for butterfly pupa. A cocoon is a covering made by some insects, not a synonym for every pupa.

5 I do I keep the animal name in the claim

Shared I do time. Say: This describes a European rabbit and a robin. I retain their names because one species cannot establish a universal rule. A fledgling has left the nest but is not necessarily independent or reproductively mature.

6 We do Repair three archive labels

Use movable labels or the HTML prediction/test panel. Correct order egg → tadpole → froglet → adult; reproduction then produces eggs of the next generation. Ask where the identity changes from parent to offspring.

7 We do Recover the missing stage

Shared We do time. Place pupa between larva and adult. Development occurs in this living stage. Choose evidence, test, then explain using card B; paper route gives the same evidence.

Reveal: Egg → larva → pupa → adult. The pupa is alive and developing.

8 We do Challenge the word all

Shared We do time. Grasshopper challenges all insects have a pupa. Platypus challenges all mammals give live birth. Better claims: monarch butterflies have a pupal stage; rabbits give live birth, but some mammals lay eggs. Do not change mammal classification because a species lays eggs.

9 Hinge What does the returning arrow mean?

A incorrect: one adult does not become its own egg. B correct: reproduction produces a new generation. C incorrect: stage positions do not set calendar duration. Re-teach with two different coloured tokens for parent and offspring.

A. The same adult turns into an egg. - The egg is a new individual, not the adult changing back.

B. Reproduction produces the next generation. - The arrow links parents to new offspring.

C. Every stage takes exactly the same time. - A life-cycle diagram shows order; equal spacing does not mean equal time.

10 You do Build a reliable display

Select one route. Supported pupils complete key stage words and matching; standard pupils make four sequences; stretch pupils combine the four core sequences with counterexamples. Dictated stage explanations can count as independent evidence.

11 You do Check what your arrows say

Shared You do time. Point to a single arrow and ask what changes. Prompt with the evidence card rather than giving the next stage. No requirement to copy scientific species names.

12 Review An audience checks the display

Use a paired check, teacher conference or private self-check. Look for order, named species and meaning of the reproductive arrow. Pupils need not perform publicly.

13 Review A smaller animal is not always an adult

Shared review time. Ask pupils to explain why leaving a nest or changing shape is not automatically the same as reproductive maturity. Do not add age thresholds.

14 Exit Three final archive checks

E1 pupa/chrysalis. E2 tadpole, froglet, nestling or fledgling, matched to the named species. E3 reproduction produces new individuals. Take a spoken, written or pointed response.

15 Exit Choose a useful next step

Shared exit time. Use the response to plan the scheme's following comparison lesson. A confidence choice is not scored as scientific attainment.

Answers and assessment

R1-R3

R1 offspring are young produced by parents. R2 a young bird is not yet a mature adult. R3 one species cannot establish a universal rule for every animal.

W1-W3

W1 egg → tadpole → froglet → adult. Hatching links egg and tadpole; development produces legs and the froglet body; growth and maturation lead to the adult. Reproduction produces the next generation of eggs. W2 pupa belongs between larva and adult; it is a living stage during which the body develops. W3 grasshopper has no pupa; platypus lays eggs but is a mammal. Accept narrower true statements such as "monarchs have a pupal stage" and "rabbits give live birth; some mammals lay eggs".

Hinge A-C

A incorrect: an adult does not turn into its own egg. B correct: reproduction produces new individuals. C incorrect: diagram spacing does not measure elapsed time.

S1-S4

S1 missing words in row order: tadpole; pupa; mature adult; fledgling. S2 European rabbit. S3 offspring or young. S4 examples: tadpole develops legs and absorbs its tail; kit grows into a young rabbit; nestling develops feathers and later fledges. Do not accept an adult turning back into its own egg.

N1-N5

N1 egg → tadpole → froglet → adult; reproductive link leads to new eggs. N2 egg → larva/caterpillar → pupa/chrysalis → adult; living development occurs in the pupa. N3 development inside mother → newborn kit fed on milk → growing young rabbit → mature adult; adults can produce offspring. N4 egg → nestling → fledgling → mature adult; adults can produce new eggs. N5 the circular representation joins generations through reproduction; it does not reverse one individual's age.

T1-T4

T1 use the four accurate N1-N4 sequences and state that reproduction produces new offspring. T2 grasshopper counterexample shows not every insect has a pupa; platypus counterexample shows not every mammal gives live birth. T3 equal diagram spacing gives no duration; dated observations following identified individuals under recorded conditions would provide timing evidence. T4 resemblance or leaving a nest does not establish reproductive maturity; each still grows and develops. Accept other accurate model limitations, including omitted mortality and individual variation.

E1-E3 and L1

E1 pupa or chrysalis. E2 a correctly paired example: common-frog tadpole/froglet or robin nestling/fledgling. E3 reproduction produces new individuals of the next generation. L1 is a reflection, not a test of personal disclosure.

Success and response to misconceptions

Secure: ordered cycles for all four named examples and an explanation of reproduction. Re-teach jumbled stages with movable labels. Use separate parent and offspring tokens if pupils reverse an individual's age. Test a broad group claim with one named counterexample; extra species need not be memorised.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 8; We do 8; hinge 3; You do 10; review 4; exit 2. Zero-minute slides share their stage time.

Supply shared page 1, one selected route page 2, 3 or 4, and page 5. Do not require all three routes. An independent spoken explanation can be recorded verbatim by an adult.

Interactive We do: predict a sequence, test the annotated SVG evidence, then explain or revise. Use Pause or Reset as needed. Paper W1-W3 assesses the same reasoning within the existing eight minutes.

The diagrams show selected stages and links, not anatomy, actual relative sizes or elapsed time. Some individuals die before reproducing; a cycle diagram is not a guarantee of survival.

Core assessment describes four named life cycles. Counterexamples support careful reasoning and bridge to next week's comparison outcome; their technical classification details are optional extension.

Access and responsive teaching

Supported

Use supplied stage words, a matching table and short sentence frames.

Standard

Create four accurate sequences with labelled development or reproduction arrows.

Stretch

Repair a misleading guide using named counterexamples and model limitations.

Regulation

Offer quiet sequencing, pointing, drawing, dictation or private writing. Use animal examples; pupils may pass and return to a question. Response routes follow current learning evidence.

Misconceptions to check

A life cycle means the same adult turns into a baby again.

Adults can reproduce to produce new individuals; the next generation begins.

Check: Ask whether the egg after an adult is that adult or its offspring.

Every insect has a pupa.

Butterflies have a pupa; insects such as grasshoppers develop through nymph stages without a pupa.

Check: Compare the monarch and grasshopper cards.

A pupa is dead or simply resting.

The living insect develops during the pupal stage.

Check: Ask what changes between caterpillar and adult butterfly.

Every mammal gives birth to live young.

Most mammals do, including rabbits. Egg-laying mammals include the platypus.

Check: Ask which card challenges the word every.

Word	Meaning
life cycle	The stages in a living thing's life, including reproduction that produces the next generation.
offspring	Young produced by parents.
larva	An immature stage with a different body form from the adult, such as a caterpillar.
pupa	The stage between larva and adult in insects with complete metamorphosis.
metamorphosis	A major change in body form during development.
fledgling	A young bird that has left the nest and is developing its flying skills.

Scientific references

[Froglife Common Frog](#)

Checked September 2026. Egg, tadpole, froglet and reproductive adult sequence; local timing varies.

[Monarch Joint Venture Life Cycle](#)

Checked September 2026. Monarch egg, caterpillar, pupa and adult stages. A butterfly pupa is not a silken cocoon.

[Animal Diversity Web European rabbit](#)

Checked September 2026. Development before birth and young fed with milk. Exact ages are not required in this lesson.

[RSPB Brilliant Broods](#)

Checked September 2026. Robin eggs and nests; fledglings remain young and may still receive parental care.

[Arizona State University Incomplete Metamorphosis](#)

Checked September 2026. Egg, nymph and adult sequence without a pupa; grasshopper counterexample.

[Australian Museum Platypus](#)

Checked September 2026. Egg-laying mammal with young fed on milk; exception to the all-live-birth claim.